1.2: Essential Passive Components

Introduction

Welcome to the foundational concepts of electronics! In this module, we will explore the essential passive components that form the building blocks of almost every electronic circuit: **resistors**, **capacitors**, and **inductors**. Understanding these components is crucial for anyone venturing into the world of electronics, as they dictate how current flows, how energy is stored, and how signals are filtered.

1. Resistors

Resistors are fundamental two-terminal passive electrical components that implement electrical resistance as a circuit element. In electronic circuits, resistors are used to reduce current flow, adjust signal levels, divide voltages, bias active elements, and terminate transmission lines, among other uses.

1.1. Resistance and Ohm's Law

Resistance (R) is the opposition to the flow of electric current. It is measured in **Ohms** (Ω). The relationship between voltage (V), current (I), and resistance (R) is described by **Ohm's Law**:

$$V = I \times R$$

Where:

- V is the voltage across the resistor in Volts (V)
- I is the current flowing through the resistor in Amperes (A)
- R is the resistance of the resistor in Ohms (Ω)

1.2. Resistor Color Codes

Resistors often have their resistance value indicated by a series of colored bands. This color code is a standardized system used to denote the resistance value, tolerance, and sometimes the temperature coefficient of a resistor. For a typical four-band resistor, the first two bands represent significant digits, the third band is the multiplier, and the fourth band indicates the tolerance.

Color	Digit	Multiplier	Tolerance
Black	0	10^0 (1)	
Brown	1	10^1 (10)	$\pm 1\%$
Red	2	10^2 (100)	$\pm 2\%$
Orange	3	10^3 (1k)	
Yellow	4	10^4 (10k)	
Green	5	10^5 (100k)	$\pm 0.5\%$
Blue	6	10^6 (1M)	$\pm 0.25\%$
Violet	7	10^7 (10M)	$\pm 0.1\%$
Gray	8	10^8 (100M)	
White	9	10^9 (1G)	
Gold		10^{-1} (0.1)	$\pm 5\%$
Silver		10^{-2} (0.01)	$\pm 10\%$
None			$\pm 20\%$

Example: A resistor with color bands Red, Violet, Orange, Gold.

- Red: 2
- Violet: 7
- Orange: 10^3 (1k)
- Gold: $\pm 5\%$

Therefore, the resistance is $27 \times 10^3\Omega = 27k\Omega$ with a $\pm 5\%$ tolerance.

1.3. Resistors in Series and Parallel

Series Connection

When resistors are connected in **series**, their resistances add up. The total resistance (R_{total}) is the sum of individual resistances.

$$R_{total} = R_1 + R_2 + R_3 + \cdots + R_n$$

Example: Three resistors of 100Ω, 220Ω, and 330Ω are connected in series.

$R_{total} = 100\Omega + 220\Omega + 330\Omega = 650\Omega$

Parallel Connection

When resistors are connected in **parallel**, the reciprocal of the total resistance is the sum of the reciprocals of individual resistances.

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$$

For two resistors in parallel, a simpler formula can be used:

$$R_{total} = \frac{R_1 \times R_2}{R_1 + R_2}$$

Example: Two resistors of 100Ω and 200Ω are connected in parallel.

$R_{total} = \frac{100\Omega \times 200\Omega}{100\Omega + 200\Omega} = \frac{20000}{300}\Omega \approx 66.67\Omega$

2. Capacitors

Capacitors are passive two-terminal electrical components that store electrical energy in an electric field. They consist of two conductive plates separated by a dielectric (insulating) material. The ability of a capacitor to store charge is called **capacitance**.

2.1. Capacitance and Farads

Capacitance (C) is the measure of a capacitor's ability to store an electric charge. It is measured in **Farads** (F). One Farad is a very large unit, so capacitors are often rated in

microfarads (μF), nanofarads (nF), or picofarads (pF).

The charge (Q) stored in a capacitor is directly proportional to the voltage (V) across it:

$$Q = C \times V$$

Where:

- Q is the charge in Coulombs ©
- C is the capacitance in Farads (F)
- V is the voltage in Volts (V)

2.2. Types of Capacitors

Capacitors come in various types, each suited for different applications based on their dielectric material, construction, and electrical characteristics.

Type	Dielectric Material	Characteristics	Common Applications
Ceramic	Ceramic	Non-polarized, small, low cost, wide capacitance range	High-frequency decoupling, resonant circuits
Electrolytic	Oxide layer	Polarized, high capacitance, larger size, lower frequency	Power supply filtering, audio coupling
Tantalum	Tantalum pentoxide	Polarized, high capacitance, small size, stable	Mobile phones, automotive electronics
Film	Plastic film	Non-polarized, stable, good for AC applications	Audio circuits, timing circuits, power factor correction

2.3. Capacitor Behavior

DC (Direct Current) Behavior

When a capacitor is connected to a DC voltage source, it charges up until the voltage across it equals the source voltage. Once fully charged, it acts like an open circuit, blocking the flow of DC current.

AC (Alternating Current) Behavior

When a capacitor is connected to an AC voltage source, it continuously charges and discharges, allowing AC current to flow through it. The capacitor's opposition to AC current is called **capacitive reactance** (X_C), which decreases as frequency increases.

$$X_C = \frac{1}{2\pi f C}$$

Where:

- X_C is the capacitive reactance in Ohms (Ω)
- f is the frequency in Hertz (Hz)
- C is the capacitance in Farads (F)

3. Inductors

Inductors are passive two-terminal electrical components that store energy in a magnetic field when electric current flows through them. An inductor typically consists of a conductor (like a wire) wound into a coil.

3.1. Inductance and Henrys

Inductance (L) is the property of an electrical conductor by which a change in current through it induces an electromotive force (voltage) in both the conductor itself and in any nearby conductors. It is measured in **Henrys** (H). Similar to Farads, Henrys are often expressed in millihenrys (mH) or microhenrys (μH).

The voltage (V) induced across an inductor is proportional to the rate of change of current (dI/dt) through it:

$$V = L\frac{dI}{dt}$$

Where:

- V is the induced voltage in Volts (V)
- L is the inductance in Henrys (H)
- dI/dt is the rate of change of current in Amperes per second (A/s)

3.2. Inductor Behavior

DC (Direct Current) Behavior

When an inductor is connected to a DC voltage source, it initially opposes the change in current. Once the current stabilizes, the inductor acts like a short circuit, allowing DC current to flow freely (assuming ideal inductor with zero resistance).

AC (Alternating Current) Behavior

When an inductor is connected to an AC voltage source, it continuously builds and collapses its magnetic field, opposing the change in current. The inductor's opposition to AC current is called **inductive reactance** (X_L), which increases as frequency increases.

$$X_L = 2\pi f L$$

Where:

- X_L is the inductive reactance in Ohms (Ω)
- f is the frequency in Hertz (Hz)
- L is the inductance in Henrys (H)

Practice Problems

Problem 1: Resistor Color Code

A resistor has the following color bands: Brown, Black, Red, Silver. What is its resistance value and tolerance?

Solution:

- Brown: 1
- Black: 0
- Red: 10^2 (100)
- Silver: $\pm 10\%$

Resistance = $10 \times 10^2 \Omega = 1000\Omega = 1k\Omega$ with $\pm 10\%$ tolerance.

Problem 2: Resistors in Series

Two resistors, $R_1 = 470\Omega$ and $R_2 = 1k\Omega$, are connected in series. What is the total resistance?

Solution:

$$R_{total} = R_1 + R_2 = 470\Omega + 1000\Omega = 1470\Omega = 1.47k\Omega$$

Problem 3: Resistors in Parallel

Two resistors, $R_1 = 330\Omega$ and $R_2 = 680\Omega$, are connected in parallel. What is the total resistance?

Solution:

$$R_{total} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{330\Omega \times 680\Omega}{330\Omega + 680\Omega} = \frac{224400}{1010}\Omega \approx 222.18\Omega$$

Problem 4: Capacitive Reactance

A 10μF capacitor is connected to an AC source with a frequency of 50 Hz. Calculate its capacitive reactance.

Solution:

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2 \times \pi \times 50\ \text{Hz} \times 10 \times 10^{-6}\ \text{F}} \approx \frac{1}{0.00314159}\Omega \approx 3183.1\Omega$$

Problem 5: Inductive Reactance

A $100\ \text{mH}$ inductor is connected to an AC source with a frequency of 1 kHz. Calculate its inductive reactance.

Solution:

$$X_L = 2\pi f L = 2 \times \pi \times 1000\ \text{Hz} \times 100 \times 10^{-3}\ \text{H} \approx 2 \times \pi \times 100\Omega \approx 628.32\Omega$$

References

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[3] Wikipedia. (n.d.). *Inductor*. Retrieved from https://en.wikipedia.org/wiki/Inductor